Round 4, item 01: the genus-3 relation phase, second pass

2026-10-16. Round rules: ../RULES.md (at most 3 single-threaded jobs, 4.5 GB and 60 minutes per job). Protocol, frozen before test data, with amendments D1-D2: PROTOCOL.md. Predecessor: round 3 item 01.

- Tool: `examples/ic_improve4_01_genus3_rel.rs`, a copy of round 3's instrument. `src/` is not touched.
- Analysis: `examples/ic_improve4_01_analyze.rs`.
- Figure: plot.py, which plots `summary_all.json`, `pertrial_w.json` and `tune.json` only.
- Records:
  - test: results/run/, with tables analysis.md (incumbent, C1, C3, C1+C3, C4; primary), analysis_c2.md and analysis_c2pair.md (C2), analysis_aa.md (A/A and the W-walk control), analysis_c4dec.md (C4 decomposition), pertrial_r2.md and pertrial_w.md (cost per trial);
  - gate: gate0.md; reproduction: results/repro/;
  - training: results/train/ (wtune.md, tune.md).

Verdict

Incumbent: round 3's **R2**, `r2:lpsgebw2`: round 3's R1 relation primitives (one-inversion Cantor addition, Stickelberger pre-filter, root finding reusing `x^((q-1)/2) mod u`) with kappa_R2(b) = 1.306 × 0.9102^(b-20), on `lpsgebw2` (large primes as columns, SGE, block Wiedemann (2, 1)).

Design: held-out curves 0-2 at q = $2^{12..2}22$ and curves 3-4 at $2^{18..2}21$; paired on instance and target (and on the walk for C1 and C3); unit `M_q + ceil(log2 q) I_q + M_ell`; 2000 paired bootstrap resamples.

Floors. A/A (`aa:lpsgebw2`, an independent single walk, 12..21): exponent Delta -0.0012 [-0.0037, +0.0014], so eps_e = 0.0037; ratios 0.994-1.023 (1.023 at 2^12, <= 0.006 from 2^13 up), so eps_c = 0.023.

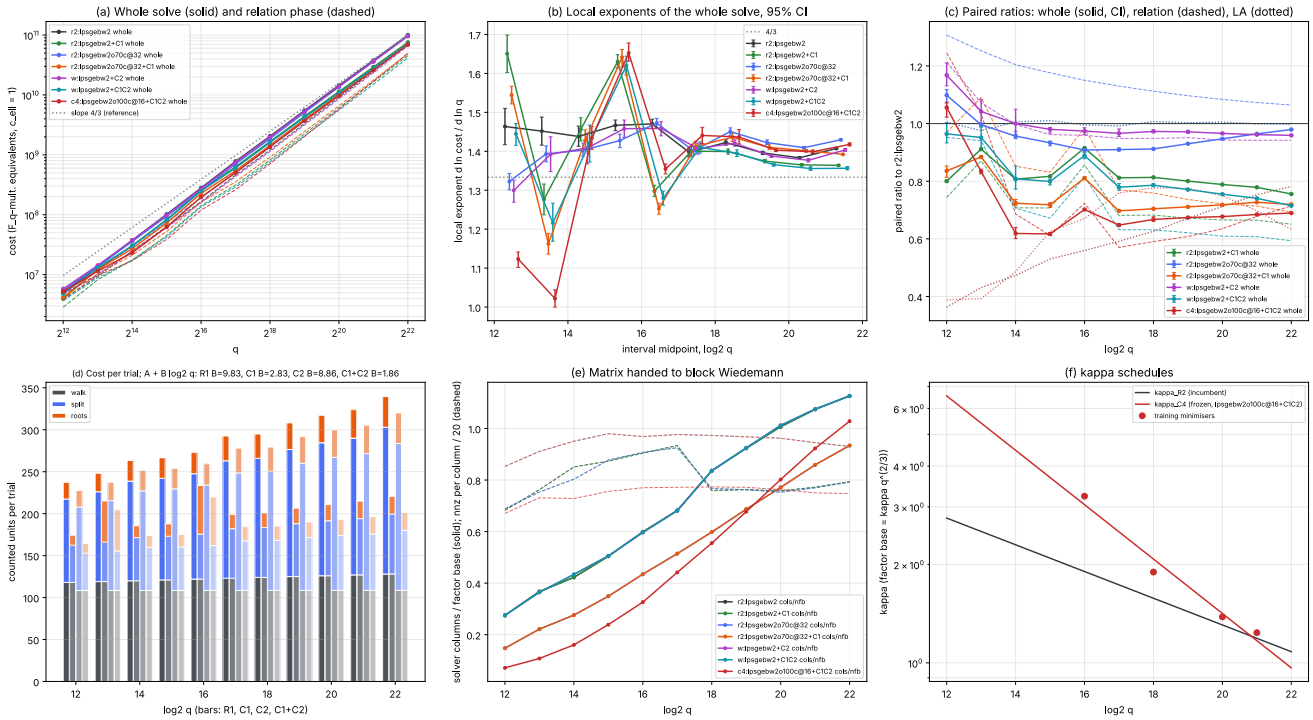
candidate	exponent, whole: Delta [95%], 12..22 and 16..22	constant: paired ratio at 2^22 [95%]	other levels	verdict
incumbent `r2:lpsgebw2`	slope 1.430 [1.428, 1.432] (12..22), 1.408 (16..22)	1 (9.967e10)	local exponent 21-22: 1.408	-
C1: field-level splitting test (same relations)	**-0.0152 [-0.0154, -0.0149]**; 16..22 **-0.0341 [-0.0343, -0.0339]**	**0.756 [0.755, 0.756]**; 0.779 at 2^21	primitive: splitting test + root finding per trial **0.438** at 2^22 (211.6 to 92.7 units); per-trial log slope **B 9.75 to 2.71** per bit; local 21-22 **1.364**; trend difference **-0.0094 [-0.0097, -0.0092]**	**ADVANCES** (exponent, constant, primitive); moves toward 4/3
C2: batched walks, W = 32	-0.0187 [-0.0221, -0.0154]*; 16..22 -0.0032 [-0.0054, -0.0011] (inside eps_e)	**0.959 [0.959, 0.959]** against the incumbent; **0.960** against its own unbatched twin (exactly paired)	primitive: walk addition per trial **0.848** (128.0 to 108.6 units at 2^22); B 9.83 to 8.86	**ADVANCES** (constant, primitive); exponent MATCHES
C3: oversample 70%, clique removal, SGE cx 32 (`lpsgebw2o70c@32`, same walk)	-0.0089 [-0.0102, -0.0073]; 16..22 **+0.0203 [+0.0194, +0.0211]**	**0.979 [0.979, 0.980]**; 0.908 at 2^16, 1.099 at 2^12	LA 0.782 and relation 1.064 at 2^22; solver columns 26277 vs 31695; local 21-22 1.429; trend difference +0.0156	**TRADES** (constant ADVANCES; phases opposite; gain shrinks with q)
C1 + C3 (same walk)	-0.0219 [-0.0233, -0.0203]; 16..22 -0.0106 [-0.0115, -0.0097]	0.719 [0.719, 0.720]	local 21-22 1.392; trend difference +0.0071	ADVANCES (constant, exponent)
C1 + C2 (W = 32, kappa_R2)	-0.0380*; 16..22 **-0.0410 [-0.0432, -0.0388]**	**0.715 [0.715, 0.715]**	local 21-22 **1.356**; trend difference +0.0046*	ADVANCES (constant; exponent over 16..22)

candidate	exponent, whole: Delta [95%], 12..22 and 16..22	constant: paired ratio at 2^22 [95%]	other levels	verdict
C4: C1 + C2 + C3 (`lpsgebw2o100c@16`), kappa re-tuned	-0.0312 [-0.0331, -0.0290]*; 16..22 **+0.0045 [+0.0035, +0.0055]**	**0.690 [0.689, 0.690]**; 0.685 at 2^21	LA 0.632, relation 0.714 at 2^22; local 21-22 1.418; trend difference +0.0362; 1.056 at 2^12	**TRADES** (constant ADVANCES, exponent over 16..22 REGRESSES)

* The W-walk arms (C2, C1+C2, C4) carry the start-up cost of 31 extra walks (62 scalar multiplications), charged in full. It makes the W-walk control `w:lpsgebw2` (R1, unbatched) cost 1.199x the incumbent at 2^12 and 1.071x at 2^13, fading to 0.999x at $2^{21-2}22$. That alone gives the control an exponent Delta of -0.0169 over 12..22 but -0.0004 over 16..22 (analysis_c2.md). So for W-walk arms the 12..22 exponent mostly measures this fixed overhead, not a gain; their exponent verdicts above use 16..22 (deviation 3 below).

What this means.

- **C1 is the round's clear result.** Splitting a square-discriminant cubic needs only the cubic character of the Cardano resolvent ratio y = z2/z1, and y + 1/y = 2 + 27 disc / P'^3 lies in F_q. A Lucas ladder (2 products per bit, one inversion) replaces R1's `x^((q-1)/2) mod u` (12-15 products per bit), and the same ladder with exponent 3^{-1} mod n/3 gives a cube root for root finding. Relations are identical: decisions and root sets were asserted equal on every trial of every job.
  - Cost per trial at 2^22: 339.6 to 220.7 units (0.650). The log-dependent slope falls from 9.75 to 2.71 units per bit, so the per-trial log factor's share of the local exponent drops from 0.043 to 0.018 at 2^21.
  - This is the lever round 3's diagnosis named, and it moves the exponent: Delta -0.015 over 12..22 and -0.034 over 16..22, ratio falling every bit from 2^17 (0.812, 0.813, 0.801, 0.789, 0.779, 0.756), local exponent 1.364 at both 20-21 and 21-22 against the incumbent's 1.383 and 1.408.
  - At q = 8209 (2^13) and 65537 (2^16), 9 divides n = q - (-3|q), there is no one-exponent cube root, and root finding falls back to R1's route; the ratio there is 0.911 and 0.915.
- **C2 is a constant.** Batching the one inversion per Cantor addition over 32 lockstep walks saves about ceil(log2 q) - 3 units per trial: 0.96x whole, 0.94x relation at $2^{20-2}22$. W = 32 walks need the same number of trials as one walk (the control's ratio is 0.998-1.007 from 2^15 up).
- **C3 is a constant that shrinks with q.** Oversampling plus clique removal makes the matrix smaller (columns 0.83x at 2^22) and LA cheaper (0.78x), for 6.4% more relation work. The LA ratio rises with q (0.56 at 2^16 to 0.78 at 2^22) and so does the whole ratio (0.908 to 0.979; below 2^14 the fixed cx = 32 and oversampling cost more than they save, 1.099 at 2^12): the 2LP graph near its threshold gives cheap extra cycles, but each extra relation saves less LA at large q. It raises the LA exponent (+0.08 over 16..22) and moves away from 4/3.
- **C4 is the cheapest arm at 2^22 (0.690x) but its frozen kappa schedule regresses the exponent.** The training rule gave ln kappa_C4(b) = 0.3478 - 0.1916 (b - 20), twice as steep as kappa_R2, partly because kappa*(16) sat at the top of the grid (clamped). A falling kappa raises the exponent, as in rounds 2 and 3: C4's ratio rises from 0.617 (2^15) to 0.690 (2^22). C1 + C2 at the incumbent's kappa (0.715x at 2^22, local exponent 1.356) has the better exponent; C1 + C3 is 0.719x.
- **Distance remaining to 4/3:** fitted over 16..22, C1 + C2 1.368 (0.035 above 4/3) and C1 1.374 (0.041), against the incumbent's 1.409 (0.075). Local 21-22: C1 + C2 1.356 (0.023), C1 1.364 (0.031), incumbent 1.408 (0.075). The matched-rho gap is not measured in this item (no rho arm, as in round 3).



cost, local exponents, paired ratios, per-trial cost, matrix size and kappa schedules

(a) geometric-mean cost per size (curves 0-2; 0-4 at 18..21 for the `r2` and `c4` arms; W-walk arms curves 0-2), whole (solid) and relation phase (dashed); slope-4/3 reference. (b) local exponents with 95% CIs. (c) paired ratios against the incumbent: whole (CI), relation (dashed), LA (dotted). (d) counted cost per trial by primitive (walk, splitting test, root finding), bars R1, C1, C2, C1+C2 from the `w` stream. (e) solver columns per factor-base element (solid) and nnz per column / 20 (dashed). (f) kappa_R2 and the frozen kappa_C4 with the training minimisers. Plotted from summary_all.json (all arms on common curves).

Per-size paired ratios (whole solve) against `r2:lpsgebw2`

From analysis.md and analysis_c2.md (95% CIs there).

log2 q	incumbent cost	C1	C3	C1+C3	C4 (kappa_C4)	C2	C1+C2	W-walk control	A/A
12	4.956e6	0.800	1.099	0.836	1.056	1.168	0.964	1.199	1.023
13	1.369e7	0.911	0.996	0.884	0.833	1.042	0.952	1.071	0.995
14	3.744e7	0.807	0.956	0.724	0.619	1.000	0.808	1.029	0.995
15	1.012e8	0.817	0.933	0.718	0.617	0.980	0.799	1.009	1.005
16	2.797e8	0.915	0.908	0.811	0.702	0.974	0.889	1.003	0.994
17	7.754e8	0.812	0.910	0.697	0.648	0.966	0.779	0.995	0.995
18	2.046e9	0.813	0.911	0.704	0.666	0.973	0.786	1.003	1.004
19	5.476e9	0.801	0.930	0.711	0.673	0.971	0.771	1.004	1.004
20	1.441e10	0.789	0.947	0.718	0.676	0.966	0.755	1.000	1.002
21	3.757e10	0.779	0.965	0.727	0.685	0.962	0.741	0.999	0.997
22	9.967e10	**0.756**	**0.979**	**0.719**	**0.690**	**0.959**	**0.715**	0.999	-

The C2, C1+C2 and control columns use curves 0-2 at every size; the others use curves 0-4 at 18..21.

Exponents by phase (12..22, Delta against the incumbent; incumbent relation 1.419, LA 1.458):

arm	relation	LA	whole, `c_ell = 9` (12..22 / 16..22)
C1	-0.0299 [-0.0302, -0.0297]	0	-0.0024 / -0.0103
C3	-0.0278	+0.1031	+0.0507 / +0.0683
C1+C3	-0.0576	+0.1031	+0.0548 / +0.0590
C4	-0.0622 (16..22: +0.0231)	+0.0858 (16..22: -0.0209)	+0.0373 / -0.0148

C3 and C4 raise the LA exponent over 12..22 because their LA saving is largest at small q.

Candidates

C1 (instrument: `split_c1`, `roots_c1`). For u = x^3 + a x^2 + b x + c over F_q (every instance field is a prime field, q > 3):

step	R1 (incumbent)	C1
discriminant, Legendre symbol (all trials)	13 products + square-and-multiply	same
square discriminant (half the trials)	`x^((q-1)/2) mod u`, `x^q = x w^2`: 12-15 products per bit	P′ = 3b - a^2, t = 2 + 27 disc / P′^3 (5 products, 1 inversion), Lucas ladder V_{n/3}(t) = 2 with n = q - (-3
root finding (1/6 of trials)	monic(w -+ 1) as quadratic factor, else Cantor-Zassenhaus	V_e(t) with e = 3^{-1} mod n/3 gives gamma + 1/gamma for a cube root gamma of y; t1 = +- sqrt(-P′(V_e + 2))/3 (square root counted product by product), sign by evaluating u, then the quotient quadratic (incumbent's formula and charge)

Why it works: a squarefree cubic with square discriminant is split or irreducible; with Cardano's resolvent roots z1, z2 (z1 z2 = -P^3/27, a cube) it splits iff z1 is a cube in F_q(sqrt(-3)), iff y = z2/z1 is a cube in the cyclic group of order n (F_q^* if q = 1 mod 3, the norm-1 torus of F_{q^2} if q = 2 mod 3), iff y^{n/3} = 1, iff V_{n/3}(y + 1/y) = 2. The suggested shared exponent for the Legendre symbol and sqrt(disc) is not needed: the test uses no sqrt(disc). Fallbacks: 9 | n (b = 13, 16) and P′ = 0 use R1's root finding, charged.

C2 (`cadd_fast_pre`, `cadd_batch`, `cadd_fast_fin`). R1's addition split at its single inversion; 32 lockstep walks invert r q4 jointly (3(W-1) products + 1 inversion); each inverse is checked. Training (T-W, b = 18): batched / unbatched relation cost 0.953 (W = 8), 0.948 (32), 0.947 (128); rule gave W = 32.

C3 (`clique_filter`, oversampling thresholds). Training at kappa_R2 (b = 16, 18, 20, 21): ratio to the incumbent LA arm 0.945 (`o35c@16`), 0.937 (`o70c@16`), 0.940 (`o100c@16`), **0.933 (`o70c@32`, chosen)**.

C4. Training picked `lpsgebw2o100c@16` with C1 and C2 and kappa*(b) = 3.235 (clamped), 1.895, 1.382, 1.237 at 16, 18, 20, 21: kappa_C4 = 6.56, 3.05, 1.42, 1.17, 0.97 at 12, 16, 20, 21, 22, against kappa_R2 = 2.77, 1.90, 1.31, 1.19, 1.08. The decomposition (analysis_c4dec.md) gives C4's R1, C1 and C2 variants on the same walk.

Considered and dropped: none of C1-C4 was dropped. A Jacobi-symbol Legendre test (priced like an inversion) would cut a further ~0.5 log2 q units per trial, but that changes the price of an operation rather than the algorithm, so it was not used.

Controls and gates

- **Gate 0:** 41/41 `r2:lpsgebw2` test records (curves 0-4, 12..22) equal round 3's `r2:lpsgebw2+R1` in k, cost, relation cost, LA counters and steps (gate0.md).
- **Shadow equality (C1):** asserted on every trial of every job (145 test, 60 training); 0 aborts.
- **Batched inverses (C2):** `r q4 z = 1` asserted on every batched addition; 0 aborts.
- **A/A:** Delta -0.0012 [-0.0037, +0.0014] covers 0.
- **Correctness:** every logarithm verified (`k phi(G) = phi(T)`) and equal to the planted k: `r2` 41 records x 4 arms, `c4` 41 × 4, `w` 33 × 4, `aa` 30 × 2; 0 failures (analysis.md header).
- **Reproduction:** the rebuilt binary reproduced two earlier records exactly (results/repro/).

Deviations and limits

1. **Container restart** during training (about 00:20): the three b = 21, kappa x 1.7 jobs were killed and rerun; nine finished b = 21 records were kept. The binary was rebuilt from the final source (doc comments and `rustfmt` only) and reproduced earlier records exactly; hashes in env.txt. Amendment D2.
2. **Scheduler bug** (23:40-00:03): the slot counter also matched the `timeout` wrapper, so only 2 jobs ran; no record affected.
3. **Exponent lens for W-walk arms:** the protocol's primary exponent is 12..top. For C2, C1+C2 and C4 that range is dominated by the charged start-up of 31 extra walks (control Delta -0.017 over 12..22, -0.0004 over 16..22), so their exponent verdicts use 16..22. This was decided after seeing the control, and is stated as such. By the 12..22 rule alone, C2 and C4 would read ADVANCES on the exponent.
4. **C3's verdict** uses the protocol's TRADES clause (phases in opposite directions); by the 12..22 rule alone its exponent would read ADVANCES, while over 16..22 it REGRESSES.
5. **kappa_C4 extrapolates** below 16 and above 21, and kappa*(16) was clamped at the grid edge.
6. **Machine limit at $2^{23.}$** $^{2}22$ jobs took 14-17 minutes but peaked at 3.66-3.86 GB resident, close to the 4.5 GB address-space cap. At the measured doubling per bit, 2^23 needs about 7.5 GB, so it was not run. Memory, not time, is the limit.
7. Wall times are descriptive only; claims use counted operations. Large-prime bookkeeping and the filter are not charged (as the incumbent's), host L0.

Reproduce

```
        cargo build --release --example ic_improve4_01_genus3_rel --example ic_improve4_01_analyze
        cd research/ic_improve4_20261015/01-genus3-relations
        A=../../../target/release/examples/ic_improve4_01_analyze
        SLOTS=3 sh run_slots.sh queue_tw.txt              # T-W (relation only, b = 18)
        $A wtune results/train > results/train/wtune.md
        SLOTS=3 sh run_slots.sh queue_train.txt           # T-C3/C4 grid
        $A tune results/train results/train/tune.json > results/train/tune.md   # -> d2.env
        sh make_test_queue.sh > queue_test.txt && SLOTS=3 sh run_slots.sh queue_test.txt
        R=results/run
        $A gate $R ../../ic_improve3_20261015/01-genus3-relations/results/run > $R/gate0.md
        $A analyze $R 12 22 16 $R/summary.json r2:lpsgebw2 r2:lpsgebw2+C1 r2:lpsgebw2o70c@32
r2:lpsgebw2o70c@32+C1 c4:lpsgebw2o100c@16+C1C2 > $R/analysis.md
        $A analyze $R 12 21 16 $R/summary_aa.json r2:lpsgebw2 aa:lpsgebw2 w:lpsgebw2 > $R/analysis_aa.md
        $A analyze $R 12 22 16 $R/summary_c2.json r2:lpsgebw2 w:lpsgebw2 w:lpsgebw2+C2 w:lpsgebw2+C1C2 >
$R/analysis_c2.md
        $A analyze $R 12 22 16 $R/summary_c2pair.json w:lpsgebw2 w:lpsgebw2+C2 > $R/analysis_c2pair.md
        $A analyze $R 12 21 16 $R/summary_c4dec.json r2:lpsgebw2 c4:lpsgebw2o100c@16 c4:lpsgebw2o100c@16+C1
c4:lpsgebw2o100c@16+C2 c4:lpsgebw2o100c@16+C1C2 > $R/analysis_c4dec.md
        $A pertrial $R r2:lpsgebw2 $R/pertrial_r2.json > $R/pertrial_r2.md
        $A pertrial $R w:lpsgebw2 $R/pertrial_w.json > $R/pertrial_w.md
        $A analyze $R 12 22 16 $R/summary_all.json r2:lpsgebw2 r2:lpsgebw2+C1 r2:lpsgebw2o70c@32
r2:lpsgebw2o70c@32+C1 w:lpsgebw2+C2 w:lpsgebw2+C1C2 c4:lpsgebw2o100c@16+C1C2 > $R/analysis_all.md
        python3 -I plot.py $R/summary_all.json $R/pertrial_w.json results/train/tune.json cost_vs_q
r2:lpsgebw2 r2:lpsgebw2+C1 r2:lpsgebw2o70c@32 r2:lpsgebw2o70c@32+C1 w:lpsgebw2+C2 w:lpsgebw2+C1C2
c4:lpsgebw2o100c@16+C1C2
        sh make_pdf.sh
```

The smoke and probe records in results/smoke/ are in no fit.